

Qn	Suggested MS
1)a)	If sphere A is to move in the opposite direction, speed of sphere B will have to be higher than the initial speed of sphere A
	which will mean that total kinetic energy after collision has increased, which is not possible
b)i)	$m_{2.5}u_{2.5} + 0 = m_{2.5}v_{2.5} + m_{2.0}v_{2.0}$ $2.5(3.6) + 0 = 2.5v_{2.5} + 2.0v_{2.0}$ $9 = 2.5v_{2.5} + 2.0v_{2.0}$
	Relative speed of approach = relative speed of separation $3.6 = v_{2.0} - v_{2.5}$
	$9 = 2.5v_{2.5} + 2.0(3.6 + v_{2.5})$ $v_{2.5} = \frac{9 - 7.2}{4.5} = 0.4$
ii)	$F_{ave, 2.5kg} = \frac{\Delta p}{t} = \frac{p_f - p_i}{t} = \frac{2.5(0.4) - 2.5(3.6)}{0.5} = 16.0N$
	Using N3L, the force on the 2.0 kg crate is also 16.0 N
c)i)	$KE_i + gpe_i = KE_f + gpe_f + WD_f$
	$\frac{1}{2}(3.0)(1.5)^2 + (3.0)(9.81)(2.0) = \frac{1}{2}(3.0)v^2 + 0 + 15.0\left(\frac{2.0}{\sin 30^\circ}\right)$
	$3.4 + 58.9 = 1.5v^2 + 60$ $v = 1.24ms^{-1}$
ii)	The crate will reach the bottom of the slope with a much higher speed
	The will result in it experiencing a much larger vertical force as it moves from the slope onto the vertical surface, potentially damaging the tomatoes.